Engine Numerical Loss Re-Rolls but Does Not Degrade: An Exact-Baseline Study of LLM Failure Symptoms

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Engine and experiments: https://github.com/wwoosshh/GeulPU
Language and compiler: https://github.com/wwoosshh/geul-lang

Abstract

Large language models sometimes produce familiar failures: endless repetition, text in an unrelated script, broken spelling, wrong answers. One hypothesis attributes part of these failures to the numerical loss of inference engines—activations re-quantized to 8 bits, half-precision attention and caches, reduction orders that change with the batch. Testing it requires separating two effects that look alike in aggregate: loss that merely *re-rolls* near-tie decisions, and loss that systematically *tilts* the model toward abnormal tokens. We use GeulPU—an inference engine whose arithmetic follows a written precision contract, is bit-identical to a CPU reference, and produces identical bits across GPU generations and vendors—as an exact, noise-free baseline, and compare it with llama.cpp and with a single-loss variant on Qwen2.5-7B-Instruct and DeepSeek-R1-Distill-Qwen-7B (Q8_0, greedy decoding, identical token inputs). On 696 Korean prompts and 600 math problems, llama.cpp changes 89–100% of the outputs, yet failures appear as often as they disappear (text in another script: 36 new vs. 35 gone; wrong answers: 25 vs. 28). At 10,817 fragile steps—where an abnormal token competes closely with the chosen one—we find no tilt toward other-script or byte-split tokens and a small tilt toward repetition (+0.0039 nat, about 0.4% in odds) that is specific to repeating tokens. Under exact computation the model itself produces most failures (32% of long-form Korean answers loop to the length limit) and amplifies single slips: one forced other-script token is followed by more other-script text in 80% of continuations, against 34% without it. All hypotheses, statistics and decision rules were pre-registered.

1 Introduction

Users of large language models (LLMs) know a small set of recurring failures: the model repeats a phrase until it runs out of tokens, switches to an unrelated language or script in the middle of an answer, misspells a word it wrote correctly a moment ago, or loses the point of a problem. These failures are usually attributed to the model—its training data, its size, the decoding settings. A competing hypothesis is that part of them comes from the arithmetic of the inference engine. Production engines trade precision for speed in many places: llama.cpp re-quantizes activations to 8 bits before its quantized matrix multiplications, computes attention and stores the key–value cache in half precision, and, like most engines, changes the order of floating-point reductions with the batch composition. Each of these choices perturbs the logits slightly, and a slightly different logit at a near-tie can send a generation down a different path.

That engine numerics change outputs is well established. Changing the batch size, the number of GPUs or the engine alone moves benchmark scores by several points (Yuan et al., 2025; Pape et al., 2026), prefix caching makes the same request diverge (Patodiya, 2026), and deterministic modes have been built to remove the variation on one machine (He et al., 2025; Qin et al., 2026).

What is not established is whether this variation *degrades* the output—produces more failures—or only *re-rolls* it, turning some answers worse and as many better. The two are indistinguishable in a single run; they differ in the balance of failures that appear and disappear, and, at the level of single decisions, in whether the perturbation is biased toward abnormal tokens.

Measuring that bias needs a baseline without engine noise. If the reference itself is noisy, selecting the decisions where an abnormal token is close to the chosen one already depends on the noise, and the measured shift regresses toward zero. We therefore use as the baseline an engine whose loss is limited to what a written contract specifies and whose bits do not depend on batch, path, GPU or vendor (Section 3), feed identical token ids to every engine, and pre-register the analysis.

Our contributions are:

1. A paired design with an exact baseline that separates *re-rolling* (symptoms appear and disappear in equal numbers) from *tilting* (a systematic shift toward abnormal tokens), with two pre-registered primary tests: trajectory-level McNemar tests and a step-level tilt test at fragile steps.
2. Evidence, for two 7B models at Q8_0, that llama.cpp's numerical loss re-rolls but does not increase failures, and that its only detectable tilt is toward repetition and is about 0.4% in odds.
3. Model-level measurements under exact computation: high failure base rates, self-amplification of a single other-script slip, no effect of a single injected anomaly in the previous turn, and the fraction of Korean syllables that are emitted as byte pieces.
4. The instrument itself: an open inference engine with a written precision contract whose outputs are bit-identical across NVIDIA generations, AMD and Intel GPUs and a CPU reference (Section 3, Appendix A).

2 Related work

Engine nondeterminism and evaluation. He et al. (2025) trace run-to-run differences in served LLMs to batch-dependent kernels and provide batch-invariant kernels; vLLM and SGLang ship deterministic modes based on them. Vosti specifies deterministic inference formally and verifies an engine against the specification, showing that the production modes still diverge under some execution variations (Qin et al., 2026). Yuan et al. (2025) report accuracy swings of up to 9% for DeepSeek-R1-Distill-Qwen-7B across GPU counts, GPU types and batch sizes in bfloat16; Pape et al. (2026) find that the choice of engine alone moves benchmark scores by up to 16.6 points; Patodiya (2026) shows that llama.cpp's prefix cache makes repeated requests diverge, more so at 4 bits. These works establish variance; we ask whether the variance is biased.

Bit-exact inference across hardware. A CPU emulator can predict the bits of unmodified vLLM and Transformers runs on several NVIDIA GPUs, enabling verification without changing the engine (arXiv:2606.00279, 2026). Fixed-shape reduction orders make linear layers identical across NVIDIA Ampere, Ada and Hopper (Cooper et al., 2026). Power-of-two weight scales make CUTLASS and Triton agree bitwise and expose the limits of tolerance-based conformance tests (Chen, 2026). Gensyn's REE reproduces inference across NVIDIA GPUs, Apple GPUs and CPUs with a proprietary operator library. To our knowledge no published system shows identical bits across NVIDIA and AMD GPUs with an open, written precision contract; we use such a system as an instrument rather than as the subject.

Failure modes. The Language Confusion Benchmark measures how often models answer in the wrong language (Marchisio et al., 2024a); quantization hurts non-Latin scripts most, and

automatic metrics understate the harm (Marchisio et al., 2024b). Repetition is self-reinforcing—each repetition makes the next more likely (Xu et al., 2022)—and repetition requests can make production models diverge (Nasr et al., 2023). Under-trained "glitch" tokens trigger anomalous behaviour (Land and Bartolo, 2024). We are not aware of work that injects or removes individual sources of numerical loss and measures their effect on these failures with an exact baseline.

3 The instrument: GeulPU and the geul language

Our conclusions rest on two properties of the baseline: it is a complete inference engine that runs the same quantized model files as llama.cpp on the same GPU, and its arithmetic is exact with respect to a written contract. Both are unusual for software that is not built on Python, PyTorch or the CUDA toolkit, and the language it is written in is little known. This section describes the stack from the bottom up; every number is taken from the two public repositories.

3.1 The geul language and its compiler

geul (글, "writing") is a systems programming language whose grammar follows Korean: verbs come last, and the role of each argument is marked by a case particle instead of its position. A function is declared with its roles, `[문자열 원본을 문자열 대상에 복사]` ("copy a string *source* to a string *destination*"), and called by role, `원본을 대상에 복사하다.`; the particles 을/를 (object), 에 (destination), 에서 (source), 로 (means) and 와/과 (companion) bind the arguments, and a call whose same-typed arguments carry no particle is a compile error. Types are ordinary—`정수` is a 64-bit integer, `짧은실수` a 32-bit float, `T 참조` a pointer—and memory is managed by hand.

The compiler is self-hosted: it is written in geul (about 10,000 lines) and compiles itself. A separate reference implementation in Python (about 7,000 lines) is kept as the bootstrap seed and as a second implementation; the build checks a fixpoint—the self-hosted compiler built by the reference implementation, then by itself, then by that output, is byte-identical each time. Users need neither: the released compiler `geulc.exe` (geul 1.3.1, October 2026, MIT license; a 182 KB zip, also packaged for Windows as the PyPI package `geul`) builds everything described below. The compiler has its own complete back end—lexer, parser, semantic analysis, a typed intermediate representation, an inliner, an x86-64 code generator with linear-scan register allocation, a hand-written instruction encoder and PE32+/ELF64 writers—and uses no C compiler, assembler, linker or LLVM. A Windows executable it produces imports only `kernel32.dll` (and `shell32.dll` for the command line); the Linux target (on the research branch) needs only `libc.so.6`. Builds are reproducible byte for byte.

3.2 From geul source to GPU instructions

GPU support lives on a research branch of the compiler. A GPU kernel is an ordinary geul function; GPU features are reached through declared intrinsics: thread and block indices, shared memory, barriers, warp shuffles, asynchronous copies (`cp.async`), integer tensor-core matrix multiply–accumulate (`mma.sync` ... `m16n8k32` ... `s8.s8.s32` and `m16n8k16`), `dp4a`, and fused multiply–add, which is emitted only where the source writes it. Two back ends, both written in geul on top of the self-hosted front end, translate such kernels:

- **PTX back end** (2,574 lines)—emits PTX, NVIDIA's documented virtual instruction set. Every floating-point operation carries an explicit rounding mode (`.rn`), so the driver cannot fuse a multiply and an add on its own; approximate instructions and floating-point tensor cores, whose summation rounding the PTX specification leaves open, are not used.
- **SPIR-V back end** (4,006 lines)—emits SPIR-V 1.6 compute shaders for Vulkan (AMD, Intel and NVIDIA). Kernel parameters become push constants and global memory becomes 64-bit device addresses; division and square root are written out to be correctly rounded,

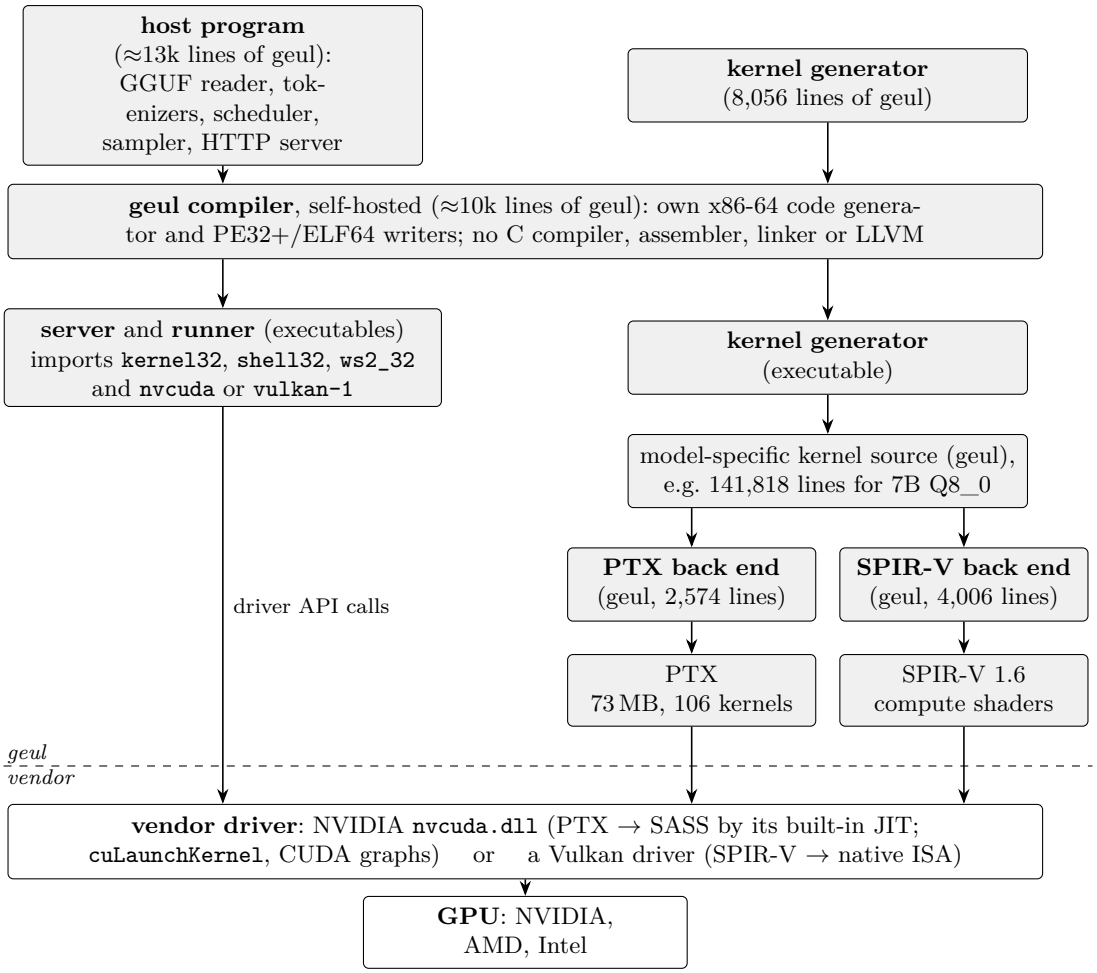


Figure 1: The GeulPU stack. Everything above the dashed line is written in geul and compiled by the geul compiler; below it is the GPU vendor's driver.

integer matrix products use cooperative-matrix instructions where the device offers them and subgroup operations otherwise.

At run time the engine hands PTX text to the NVIDIA driver (`cuModuleLoadData`), whose built-in just-in-time compiler translates it to the GPU's native machine code (SASS), and launches kernels with `cuLaunchKernel` and CUDA graphs (Figure 1). On Vulkan it creates shader modules and compute pipelines and records dispatches; a thin layer maps the driver-API calls the engine uses onto Vulkan, so the engine code is shared. Nothing from the CUDA toolkit—`nvcc`, `ptxas`, `cudart`, cuBLAS—or from a Vulkan SDK is used to build or run the engine; the continuous-integration job installs `ptxas` only to check register spills. This is the same layer at which other engines meet the GPU: llama.cpp ships kernels compiled ahead of time with `nvcc` and calls cuBLAS, vLLM runs PyTorch, CUDA and Triton kernels; GeulPU generates its own kernels and talks only to the driver. The vendors' closed compilers below PTX and SPIR-V are outside the contract, which is why bit-identity across devices is established empirically (Appendix A) rather than proved.

3.3 The kernel generator and the engine

A kernel generator, itself a geul program (8,056 lines), writes the kernel source specialized to a model's shape and weight format. For Qwen2.5-7B at Q8_0 it writes 141,818 lines of geul, which the PTX back end turns into 73 MB of PTX holding 106 kernels and 6,488 tensor-core instructions; the module is built on the user's machine on first use, and the CI regenerates 18

such modules and compares their SHA-256 hashes with the recorded ones.

The host—about 13,000 lines of geul (15,000 with its test programs)—is a complete serving engine:

- **Models**: reads GGUF files with Q8_0, Q4_0, Q4_K, Q5_K and Q6_K weights (including the Q4_K_M and Q5_K_M mixes); architectures qwen2, qwen3 and llama—Qwen2.5, Qwen3, DeepSeek-R1-Distill-Qwen, Llama 3.1 and 3.2, Mistral v0.3—with head size 128 (Qwen2.5-0.5B only through the command-line runner).
- **Text**: byte-level BPE with Unicode NFC, the Llama 3 BPE and SentencePiece tokenizers, written in geul, which match llama.cpp token for token on 2,000 lines of test text; chat templates checked against the GGUF's Jinja templates.
- **Serving**: an OpenAI-compatible HTTP server (chat and text completions, streaming, `/tokenize`), up to 8 slots with continuous batching and per-slot prompt reuse, a decode step captured as one CUDA graph, sampling and log-probabilities computed under the same written contract.
- **Footprint**: the CUDA server imports only `kernel32`, `shell32`, `ws2_32` and `nvcuda` (43 driver-API functions); the Vulkan server imports `vulkan-1` (48 functions) instead of `nvcuda`. The CI checks these import tables.

On an RTX 4070 Ti, its decoding speed is within ±5% of llama.cpp for Qwen2.5-7B and Llama-3.1-8B at Q8_0 and Q4_K_M (0.97–1.05× the time per token, depending on the llama.cpp build and the context length); prompt processing is faster than llama.cpp below about 128 tokens and up to 1.3–1.4× (Q8_0) and 1.4–1.6× (k-quant formats) slower at 8,192 tokens. Not everything is in geul: the CPU reference of the contract, the test harnesses and the benchmarks are Python (NumPy), and the optional desktop app's interface is JavaScript. The engine is Windows-first (Linux builds were tested under WSL2), uses one GPU, and needs compute capability 8.0 or newer on the CUDA path.

3.4 The precision contract

GeulPU's arithmetic follows a written contract (`docs/contract.md`) that leaves the hardware no discretion:

- **Linear layers.** Each block of 32 activations is scaled by a power of two from its largest value and rounded to integers of at most 2^{14}, stored as two signed 8-bit digits. The products with the model's integer weights are summed exactly on int8 tensor cores (prompts) or with `dp4a` (decoding)—exact integer sums do not depend on summation order—then rounded once per block and accumulated with fused multiply–adds in a fixed order.
- **Attention.** Queries, keys and values are converted the same way and the key–value cache stores the digits; scores are exact integer sums rounded once; the exponential is a written polynomial; keys are processed in fixed 64-key chunks and folded in a fixed two-level order.
- **Everything else.** Normalizations use fixed reduction trees, rotary tables are correctly rounded, and one rule fixes how subnormal numbers flush on every path (PTX `.ftz`, the Vulkan drivers and the CPU reference agree on it); sampling and log-probabilities are computed in float64 in a fixed order.

Because nothing depends on how the work is split, the bits do not depend on batch, slot, chunking, prompt-versus-decode path, GPU or vendor, and they equal those of a CPU reference implementation of the contract in NumPy; this is checked on kernel test cases, on whole models end to end, and across machines (Appendix A). The contract is close to exact: on Qwen2.5-0.5B (Q8_0) the Kullback–Leibler divergence of its next-token distributions from an fp64 computation is 2.24×10^{-6}, against 1.38×10^{-3} for llama.cpp; for 7B and 14B the corresponding ratios

(1/3,805 and 1/4,467) come from a simulation of the rounding points rather than from the engine. Exactness has a price: at equal optimization, keeping two activation digits instead of one makes prompt processing 1.26–1.38× slower (a lower bound measured by switching the tile kernels), while decoding, which is memory-bound, is unaffected.

4 Method

4.1 Arms and inputs

- **G (exact baseline)**—GeulPU under its precision contract (Section 3): loss only where the contract writes it, no dependence on batch, slot or execution path. OpenAI-compatible server, 8 slots.
- **L (production engine)**—llama.cpp build b11669 (Windows CUDA 13.4), `llama-server` with default settings (flash attention *auto*, F16 key–value cache, prompt cache on), 8 parallel slots, the same GGUF files.
- **V (one loss)**—G with a single change: the input activations of all linear layers rounded to one digit instead of two (block grid 2^{E-5}, $|n| \leq 64$—the same class as llama.cpp's 8-bit activation re-quantization); attention and everything else unchanged.

All prompts were rendered once with llama.cpp's `/apply-template` (the chat template stored in the GGUF) and tokenized with `/tokenize`; every arm received the same token ids. Decoding was greedy (temperature 0) with the same maximum length and end tokens, so the arms differ only in arithmetic. The context was 4,096 tokens per slot. Models: Qwen2.5-7B-Instruct (Korean prompts) and DeepSeek-R1-Distill-Qwen-7B (math; R1-Distill below), both Q8_0, so the weight loss is identical across arms and small.

4.2 Prompt sets

Table 1 lists the prompt sets: natural prompts, and trigger sets built to make each symptom frequent enough to study.

4.3 Symptoms

All symptoms are decided automatically from token ids and text:

- **S1 endless repetition**—the generation hit the length limit and its tail is an exact repetition: the smallest period $p \leq 512$ tokens whose repetition covers at least $\max(3p, 64)$ final tokens.
- **S2 other-script text** (Korean sets)—S2a: a run of two or more letters that are neither Hangul nor Latin (Han, kana, Cyrillic, Thai, Arabic, Devanagari, ...; Greek excluded), outside code, not present in the question; or S2b: less than 50% of the letters are Hangul.
- **S3 unfinished / S4 wrong answer** (math)—S3: length limit without `\boxed{}`; S4: the last boxed number differs from the gold answer (S3 counts as wrong).
- **S6 broken spelling** (copy trigger)—the copied span differs from the source by at least one character, or contains a broken UTF-8 character.

4.4 Fragile steps and tilt

At each generated step of G, let a be the chosen token (greedy: G's top-1) and b the highest-ranked abnormal candidate of a given type among G's top 20:

Table 1: Prompt sets.

Set	Items	Max new	Model	Source
Natural: LCB Korean	396	1,024	Qwen2.5	Language Confusion Benchmark (Marchisio et al., 2024a), Korean native and cross-lingual prompts (Apache-2.0); 3 overlong prompts excluded
Natural: long-form Korean	300	2,048	Qwen2.5	60 topics × 5 forms ("write 2,000 characters or more")
Natural: math	600	3,200	R1-Distill	gsm8k test (Cobbe et al., 2021), first 600, with "Please reason step by step, and put your final answer within `\boxed{}`."
Trigger: rare syllables	200	512	Qwen2.5	"Copy the following text exactly": 80–200-character texts with ≥ 6 syllables that the tokenizer splits into byte pieces
Trigger: mixed script	200	1,024	Qwen2.5	Korean questions about Chinese/Japanese words and names written in their own script
Trigger: repetition	200	2,048	Qwen2.5	requests to repeat a word or sentence, count in Korean words, continue a pattern
Contamination dialogs	100 × 4	512 + 1,024	Qwen2.5	two-turn dialogs; the first answer is G's own, unchanged or with one injected anomaly

- **O (other script)**—a is Hangul, Latin, digit or symbol; b contains a letter of another script (byte-piece tokens are classified by the script their lead byte implies);
- **R (repetition)**—appending a does not copy an earlier 8-token sequence of the answer, appending b does;
- **C (byte-split character)**—the step is inside a multi-byte character (earlier tokens left the UTF-8 sequence incomplete), and b completes a different character or breaks the encoding.

A step is *fragile* if the margin $m = \ln p_G(a) - \ln p_G(b)$ is below $\ln 10$. For an arm $X \in \{L, V\}$, the tilt at the step is

$$\Delta_X = \big[\ln p_X(b) - \ln p_X(a)\big] - \big[\ln p_G(b) - \ln p_G(a)\big], \tag{1}$$

computed on the same prefix (G's answer). Zero-mean numerical noise gives $\mathbb{E}[\Delta] = 0$; a systematic shift toward abnormal tokens gives $\Delta > 0$. As a control we compute the same quantity for the highest-ranked *normal* alternative at the same step. The mean Δ is estimated over all fragile steps with a 95% interval from a bootstrap that resamples prompts (10,000 resamples, seed 40); the three types are corrected with Holm's method. Decision labels (pre-registered): *tilts toward abnormal* (mean > 0, corrected $p < 0.05$), *tilts away*, *no tilt* (corrected $p \geq 0.05$ and the interval within ± 0.05 nat), *undecided* otherwise or with fewer than 200 steps.

To obtain L's distribution at G's prefixes we drive llama.cpp through its C API: the prompt is evaluated in one batch, as the server does, and G's answer tokens are then fed one at a time, eight sequences per batch, which is the server's generation path. We validated this against `llama-server` itself (Section 5.1).

Table 2: Primary trajectory tests, llama.cpp (L) against the exact baseline (G). G / L: items with the symptom in each arm; new: items with the symptom only in L; gone: only in G.

Test	Items	G / L	New	Gone	Holm p	Label
H1: Qwen2.5-7B, natural Korean, other script (S2)	696	94 / 95	36	35	1.0	shakes but does not increase
H2: R1-Distill-7B, gsm8k, wrong answer (S4)	600	80 / 77	25	28	1.0	shakes but does not increase

4.5 Trajectories, propagation and contamination

- **Trajectories (primary).** For each prompt, a symptom present only in the arm is *new*, present only in G *gone*. H1: Qwen2.5-7B, natural Korean sets, S2. H2: R1-Distill, math, S4. Exact McNemar tests, Holm over the two. Labels: *increases*, *decreases*, *shakes but does not increase* (new + gone ≥ 40, corrected $p \geq 0.05$), *undecided* (new + gone < 40; H2 would first add the next 600 problems once—not needed).
- **Propagation.** At up to 100 fragile steps per type (seed 40) we force b in G and continue for 256 tokens, and compare with G's own continuation after a: does a symptom of the same type appear again?
- **Contamination.** We insert one anomaly into G's first answer of a two-turn dialog—a Chinese word, a misspelled word, or a duplicated sentence—and compare the second answer with the one generated after the unchanged first answer.

4.6 Pre-registration

The design, the statistics and the decision rules were committed before any measurement (commit `9190f76`); the prompt sets and tools before the main runs (`d88b6cf`, `eedb304`); deviations are listed in Appendix B.

5 Results

5.1 Validity checks

- **Identical inputs**: in every arm and item, the number of prompt tokens the engine received equals the number sent.
- **Determinism**: 50 prompts re-run at concurrency 1 give identical tokens in G and in V; 50 forced-token continuations re-run give identical tokens.
- **The L measurement path**: for 50 prompts generated by `llama-server` with one slot, the follower's argmax equals the server's token at all 3,358 steps up to the first divergence from G.
- **Human check of the symptom judges** (20 items per symptom: 10 judged positive, 10 negative): repetition 19/20, other-script 17/20—below the pre-registered 90%. The other-script judge counted Han characters used to explain a Chinese word and English song titles quoted from the prompt. Because both arms of each comparison use the same judge, the paired tests are not biased toward either arm; a stricter post-hoc judge (Section 5.6) leaves every conclusion unchanged.

Table 3: Tilt at fragile steps of Qwen2.5-7B (972,380 output tokens). $\bar{\Delta}$: mean tilt in nats (Eq. 1); CI: 95% prompt-cluster bootstrap interval; control: the same quantity for the best normal alternative at the same steps; last row: steps at which llama.cpp itself chose b.

	O: other script	R: repetition	C: byte-split character
Fragile steps	3,042	4,975	2,800
per 1k tokens	3.13	5.12	2.88
$\bar{\Delta}_L$	-0.00006	$+\mathbf{0.0039}$	-0.0034
95% CI	$[-0.0046, +0.0046]$	$[+\mathbf{0.0015}, +\mathbf{0.0063}]$	$[-0.0074, +0.0004]$
Holm p	0.99	**0.003**	0.18
Label	no tilt	**tilts toward abnormal**	no tilt
Control $\bar{\Delta}_L$	$+0.0010$	-0.0011	$+0.008$ (103 steps)
$\bar{\Delta}_V$	-0.018	$+0.012$	$+0.012$
95% CI	$[-0.031, -0.006]$	$[+0.005, +0.018]$	$[+0.002, +0.023]$
L chose b	37 (1.2%)	101 (2.0%)	55 (2.0%)

Table 4: Symptom counts of Qwen2.5-7B under exact computation (arm G).

Set	Items	S1 repetition	S2 other script	S6 broken copy
Natural: LCB Korean	396	6	65 (16%)	–
Natural: long-form Korean	300	95 (32%)	29	–
Trigger: rare syllables	200	10	44	115 (58%)
Trigger: mixed script	200	6	80 (40%)[a]	–
Trigger: repetition	200	52 (26%)	1	–

[a] 36 (18%) by the stricter judge of Section 5.6.

5.2 Trajectories: re-rolled, not degraded

Table 2 gives the primary tests. llama.cpp produced a different text for 89% of the LCB prompts, all 300 long-form prompts and 97% of the math problems; the first divergence came after a median of 66, 57 and 138 tokens. The single loss V changed even more (96–100%, median first divergence 21–61 tokens) and was equally balanced (H1′: 48 new vs. 43 gone; H2′: 25 vs. 28). No secondary comparison moved: endless repetition in long-form answers 19 new vs. 15 gone, broken copies 4 vs. 3, unfinished math 29 vs. 35 (all $p \geq 0.25$). Accuracy on the 600 math problems was 86.7% (G) and 87.2% (L).

5.3 Tilt: only toward repetition, and small

The repetition tilt (Table 3) corresponds to a 0.39% increase in the odds of the repeating token; the control at the same steps is zero, so the shift is specific to repeating tokens rather than a general flattening. llama.cpp chose the abnormal token only where the margin was below 0.5 nat (below 0.25 nat: 36 of 206 O steps, 101 of 754 R steps, 52 of 314 C steps). The single activation loss tilts about three times more toward repetition, also tilts toward byte errors (where llama.cpp shows no tilt), and tilts *away* from other-script tokens: different losses tilt in different directions. On R1-Distill's English reasoning (8,121 repetition steps) neither L nor V tilts (-0.0006, -0.0009).

5.4 Under exact computation, the model produces the failures

R1-Distill answered 86.7% of the math problems correctly; 59 of the 80 wrong answers did not finish within 3,200 tokens. On the rare-syllable trigger the exact model sometimes loops on an emoji, or invents a Chinese "user" turn and repeats a character until the limit (Table 4). In real

Korean text 11.2% (Qwen2.5) and 8.5% (Llama 3.1) of syllables are emitted across two or more byte-piece tokens; only 2,512 (Qwen2.5) and 599 (Llama 3.1) of the 11,172 Hangul syllables are single tokens.

5.5 Propagation and contamination

Forcing the abnormal token at an other-script fragile step makes more other-script text appear within the next 256 tokens in 80 of 100 continuations, against 34 after the original token (47 vs. 1 discordant, $p = 3.5 \times 10^{-13}$); 99 of the 100 forced tokens were complete characters, and excluding the first two characters of the continuation gives 77 of 99, so the effect is not a mechanical completion of a byte sequence. For repetition and byte-split steps the measure saturated (the original continuations already showed the symptom in 89% and 97% of cases). A single anomaly injected into the previous answer did not raise symptoms in the next one: other-script text 32 vs. 37 (5 new, 10 gone), broken copies 53 vs. 54, repeated sentences 7 vs. 10.

5.6 Sensitivity to the symptom judge

A stricter judge, defined after the human check and before re-computation, ignores foreign text inside quotes or brackets or followed by a Hangul gloss, and ignores Latin words that occur in the prompt. It agrees with the human labels on 19 of 20 items (the items were used to design it, so this is not an independent check). All conclusions hold: H1 31 new vs. 31 gone, H1′ 41 vs. 43, propagation 78% vs. 31%. Only the rates move: the mixed-script trigger's other-script rate falls from 40% to 18%; the LCB rate from 16% to 15%.

6 Discussion

Re-rolling, not tilting. At the precision of a mainstream engine with 8-bit weights, numerical loss behaves like re-sampling at near-ties: almost every long answer changes, and the failures it creates are matched by failures it removes. The harm is reproducibility—the symptom status of about one item in ten (71 of 696, 53 of 600) depends on the engine—not quality.

The one tilt. The repetition tilt is real and specific, and the single activation loss shows it three times more strongly, which points to activation rounding as one contributor. At 0.4% in odds it did not reach the trajectory level in our samples.

The model is the source. Under exact arithmetic the model loops on a third of long Korean answers, mixes scripts in a sixth of LCB answers and fails most rare-syllable copies; once a single other-script token is emitted, the model continues in that script most of the time. Mitigations for these failures belong to the model and the decoding procedure rather than to the arithmetic.

What an exact baseline is for. It made a 0.004-nat shift measurable and every case replayable bit for bit. The same result also bounds its practical value for users: effects small enough to need an exact baseline are too small to change what users see.

Limitations. Two 7B models of one family at Q8_0, greedy decoding, one production engine (llama.cpp) on one GPU (RTX 4070 Ti), mostly Korean prompts, automatic judges (the other-script judge agreed with a human on 85%), and only engine-level loss: we did not test larger losses such as 4-bit key–value caches, longer contexts or sampling. Propagation could not be measured for repetition and byte-split steps because the measure saturated. Closed frontier models could not be studied; the mechanisms (byte pieces, near-ties, self-amplification) are generic, but their rates there are unknown.

7 Conclusion

With an exact, noise-free baseline and identical inputs, the numerical loss of a production engine re-rolls LLM outputs without increasing their familiar failures; its only measurable bias is a 0.4% tilt toward repetition. The failures come from the model, which also amplifies its own slips. Making inference arithmetic exact buys reproducibility, not fewer failures, at this precision.

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A Bit-identity evidence for GeulPU

Within one machine. Against the CPU reference: every logit row of Qwen2.5-0.5B; 26,155,008 logits of Qwen2.5-7B Q8_0 with zero differences; 106 kernel cases for Q4_0 and 349 for the k-quants plus whole files; the Qwen3, Llama 3.x and Mistral files end to end. Across execution paths: decode graph, whole prompt, prompt in four pieces, one row at a time and repeated runs give identical logits and key–value caches; wikitext-2 log-likelihoods are bit-identical whether text is fed in chunks of 512, 128 or 1 token (llama.cpp's perplexity moves between 15.8171 and 15.8200 with its batch settings); the server returns identical tokens and logit hashes for 4 models × 12 requests × 4 schedules (llama-server changed its greedy answer in 11 of 12 cases); lm-evaluation-harness item results are identical at 1 and 8 concurrent requests and on CUDA and Vulkan; Linux (WSL2) output equals Windows output byte for byte.

Across machines. A cross-check package builds the compiler, the back ends, the server and the 7B module from source on a volunteer's PC (the build hashes are compared with the reference), downloads DeepSeek-R1-Distill-Qwen-7B Q8_0, solves 32 synthetic math problems greedily (≤3,200 tokens) and compares the texts and a 64-bit FNV hash of the full 152,064-entry logit row at every step (Table 5).

Table 5: Cross-machine check. Texts: problems whose text is identical to the reference; logit hashes: steps whose logit-row hash is identical to the reference.

PC	GPU (path)	Driver	Texts	Logit hashes
Reference	NVIDIA RTX 4070 Ti, Ada (CUDA)	616.56	reference	reference
Volunteer 1	NVIDIA RTX 5070, Blackwell (CUDA)	596.21	32/32	82,673/82,673
Volunteer 2	NVIDIA RTX 3080 Ti, Ampere (CUDA)	616.92	32/32	82,673/82,673
Reference	RTX 4070 Ti (Vulkan; two kernel variants)	616.56	32/32	82,673/82,673
Volunteer 3	AMD RX 9060 XT, RDNA4 (Vulkan; two variants)	AMD 26.8.1	32/32	82,673/82,673
Volunteer 4	AMD RX 9060 XT (Vulkan; two variants)	AMD 25.20.42.14	32/32	82,673/82,673

The 82,673 hashed steps are 82,657 generated tokens plus the 16 end-of-sequence steps. On the same reference PC, llama-server produced identical texts for 0 of the 32 problems between 1 and 8 concurrent requests and changed 6 final answers. An Intel UHD 770 integrated GPU reproduced the 0.5B model's records (larger models exceed its memory limits). Each run built the engine from the commit current at the time; later engine changes left the reference outputs unchanged and were re-verified on the reference PC only. The volunteers are acquaintances of the author running the published package, not independent replicators, and the 32 problems are synthetic.

B Deviations from the pre-registration

1. The 0.5B secondary model was dropped: the GeulPU server accepts only Qwen2-family models with head size 128.

2. The context was set to 4,096 tokens per slot (not specified in advance). Three LCB prompts too long to leave 256 new tokens were excluded from all arms; three others had their maximum length reduced to 808 tokens in all arms.
3. R1-Distill prompts were tokenized with the BOS token, as `llama-server`'s chat path does (the GGUF requests BOS; the template text does not contain it); the first tokenization without BOS was discarded before any measurement.
4. V's scoring of two very long LCB answers failed (context full); fragile steps whose tokens were outside V's top 20 (17, 1, 0) are missing from Δ_V.
5. Propagation for byte-split steps was measured only on the copy trigger, where a source text exists; propagation for repetition and byte-split steps saturated.
6. The misspelling injection failed in 8 contamination dialogs (fewer than three Hangul syllables in the middle sentence); those pairs are identical and do not contribute to the counts.
7. A selection bug (string comparison of item ids) emptied the determinism and follower-validation samples in the first run; the steps were re-run after the fix, without effect on any other number.
8. The other-script judge did not reach the pre-registered 90% agreement with the human check (Sections 5.1 and 5.6).

C Reproduction and data

- **Pre-registration and record** (Korean): 문서/40단계.md in the repository—design (commit `9190f76`, before any measurement), prompt sets and tools (`d88b6cf`), fixes before the main runs (`eedb304`), results (`727a218`), human check and the strict-judge analysis (`0a5ab97`).
- **Prompt sets**: 시험/질문40/ with sources, licenses (LCB: Apache-2.0; gsm8k: MIT) and SHA-256 hashes in 시험/질문40/출처.md; built by 재기/질문40.py (seed 40).
- **Tools**: 재기/증상40.py (symptom and fragile-step judges, statistics; 93 unit tests), 재기/모음40.py (tokenization through llama.cpp, generation in the three arms, V's scoring, contamination, forced-token continuations, determinism re-runs), 재기/따라40.py (follower of llama.cpp's distributions through its C API), 재기/셈40.py (analysis), 재기/사후40.py (strict judge; 12 unit tests), 공식실행/시험40.sh (the order of steps).
- **Command**: `bash` 공식실행/시험40.sh, given the repository, the work directory, an ASCII-path model directory and the llama.cpp directory, runs build → tokenization → G → contamination → V → L → follower → checks → fragile steps → forced-token continuations → determinism → analysis.
- **Versions and hashes** (문서/결과/40단계/40_기록.txt): GeulPU server `010f5dae...`, kernel generator `a304a884...`, modules `qwen2Q8_7b` and `qwen2Q8_7b_자리하나` equal to the repository's module-hash table; llama.cpp b11669 (commit `620d2225c`, Windows CUDA 13.4 build); models Qwen2.5-7B-Instruct Q8_0 (SHA-256 `52c5bf14...`) and DeepSeek-R1-Distill-Qwen-7B Q8_0 (`7307bfbb...`); NVIDIA RTX 4070 Ti, driver 616.56.
- **Results**: 문서/결과/40단계/40_판정.json (all numbers), 40_사후_S2엄격.json (strict judge), 사람확인.md (human check).
- **Raw data**: Zenodo, DOI 10.5281/zenodo.23417759 (Woo, 2026)—five zip archives (1.7 GB compressed, 4.3 GB unpacked): every generated token with its top-20 log-probabilities for G, V and L, the follower's per-step log-probabilities, the fragile steps, forced-token continuations, contamination dialogs, the token ids sent, and the engine executables and kernels used, with a README describing the formats and SHA-256 manifests. Every case generated by G can be replayed bit for bit on any supported GPU or on the CPU reference.